Evaluation of Dizziness

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- Consider the following during evaluation:
  - Is the dizziness dangerous (arrhythmia, transient ischemic attack, stroke)
  - Is it vestibular
  - If vestibular, is it peripheral or central

TABLE 21-1 Features of Peripheral and Central Vertigo

Sign or Symptom	Peripheral (Labyrinth or Vestibular Nerve)	Central (Brainstem or Cerebellum)
Direction of associated nystagmus	Unidirectional; fast phase opposite lesion[a]	Bidirectional (direction-changing) or unidirectional
Purely horizontal nystagmus without torsional component	Uncommon	May be present
Purely vertical or purely torsional nystagmus	Never present[b]	May be present
Visual fixation	Inhibits nystagmus	No inhibition
Tinnitus and/or deafness	Often present	Usually absent
Associated central nervous system abnormalities	None	Extremely common (e.g., diplopia, hiccups, cranial neuropathies, dysarthria)
Common causes	Benign paroxysmal positional vertigo, infection (labyrinthitis), vestibular neuritis, Ménière's disease, labyrinthine ischemia, trauma, toxin	Vascular, demyelinating, neoplasm

[a] In Ménière's disease, the direction of the fast phase is variable.
[b] Combined vertical-torsional nystagmus suggests BPPV.

Quick Approach to Dizziness

	Supine sxs	Rapid onset	Nystagmus	Comorbids
Disequilibrium	-	-	-	+
Near syncope	-	+	-	+/-
Vertigo	+	+/-	+	+/-
Psychiatric	+	+	-	+
Intoxicated	+	+	+/-	+/-

Rebecca Jeanmonod, MD, FACEP "Spinning on that Dizzy Edge"; AAFP Emergency and Urgent Care, 2013

Clinical Case #1

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- 60yo female presents complaining of “dizziness”. Her symptoms have been worsening over the past hour. No recent trauma to her head. She has significant nausea and very fearful she will fall. She reports these symptoms in any position. She had a recent upper respiratory tract infection. Physical exam demonstrates horizontal nystagmus. She can walk without assistance.

- What is the proper terminology for her condition?

Clinical Case #1

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- 60yo female presents complaining of “dizziness”. Her symptoms have been worsening over the past hour. No recent trauma to her head. She has significant nausea and very fearful she will fall. She reports these symptoms in any position. She had a recent upper respiratory tract infection. Physical exam demonstrates horizontal nystagmus. She can walk without assistance.

- What is the proper terminology for her condition?

- ***Vertigo***

Vestibular Testing

- **Fixation**
  - Test to see if nystagmus extinguishes
  - Give the patient a focal point (picture on a paper, a light, a stationary object)
  - Peripheral nystagmus WILL extinguish
  - Central nystagmus will NOT extinguish

Vestibular Testing

- **Vestibulo-ocular reflex**
  - Patient fixates on your finger
  - Turns head from side to side; eyes should maintain fixation on the object with smooth tracking
  - Ipsilateral failure with peripheral vertigo
  - Intact with central vertigo

Mnemonics Used in Practice

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- ATTEST (ACEP)
  - Steps used to determine if acute dizziness patient is experiencing is vertigo
  - Designed to improve diagnosis decrease unnecessary testing/hospitalization and reduce health care costs
  - **A**ssociated symptoms, **T**iming and **T**riggers, **E**xamination **S**igns, and **T**esting (additional tests)

- HINTS (ACEP)
  - Once diagnosis of vertigo has been established, exam to decrease neuroimaging
  - **H**ead **I**mpulse testing, **N**ystagmus, **T**est of **S**kew

Clinical Case #2

- Physical Exam
  - Nystagmus: present
  - Fixation: present
  - VOR: intact
  - Postural stability: poor with eyes open or closed
  - Hearing: unaffected

- Diagnosis?

Clinical Case #2

- Physical Exam
  - Nystagmus: present
  - Fixation: present
  - VOR: intact
  - Postural stability: poor with eyes open or closed
  - Hearing: unaffected

- Diagnosis?
  - *Stroke*

Clinical Case #3

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- Vitals normal

- Vestibular testing:
  - Nystagmus: present
  - Fixation: extinguishes nystagmus
  - VOR: abnormal
  - Postural stability: poor with eyes closed, okay with eyes open
  - Hearing: normal

- Diagnosis?

Clinical Case #3

- Vitals normal

- Vestibular testing:
  - Nystagmus: present
  - Fixation: extinguishes nystagmus
  - VOR: abnormal
  - Postural stability: poor with eyes closed, okay with eyes open
  - Hearing: normal

- Diagnosis?
  - Vestibulitis

Vestibulitis

- **Treatment**
  - Similar to Bell's Palsy
    - Corticosteroids and antivirals (Acyclovir/Valacyclovir)
  - Try to limit symptomatic treatment, as this may prolong symptoms
    - Brain eventually extinguishes input from cranial nerve VIII on the affected side

Clinical Case #4

- 40yo female presents complaining of “room spinning”. Started as she was reaching for a bowl off the top shelf. Symptoms last for less than a minute and area accompanied by nausea and emesis. No symptoms while sitting very still, but they do recur after change in position. She walks into the exam room unassisted.

Clinical Case #4

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- Vitals normal
- Nystagmus present
- Vestibular testing:
  - Fixation: extinguishes nystagmus
  - VOR: normal
  - Postural stability: bad with eyes closed, okay with eyes open
  - Hearing: normal

Clinical Case #4

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- What test would you like to attempt next?
  - Dix Hallpike Maneuver
- Dix Hallpike is positive
- Symptoms last less than 1 minute
- No movement, No symptoms
- Diagnosis?

Clinical Case #4

- What test would you like to attempt next?
  - Dix Hallpike Maneuver

- Dix Hallpike is positive
- Symptoms last less than 1 minute
- No movement, No symptoms

- Diagnosis?
  - BPPV

Benign Paroxysmal Positional Vertigo

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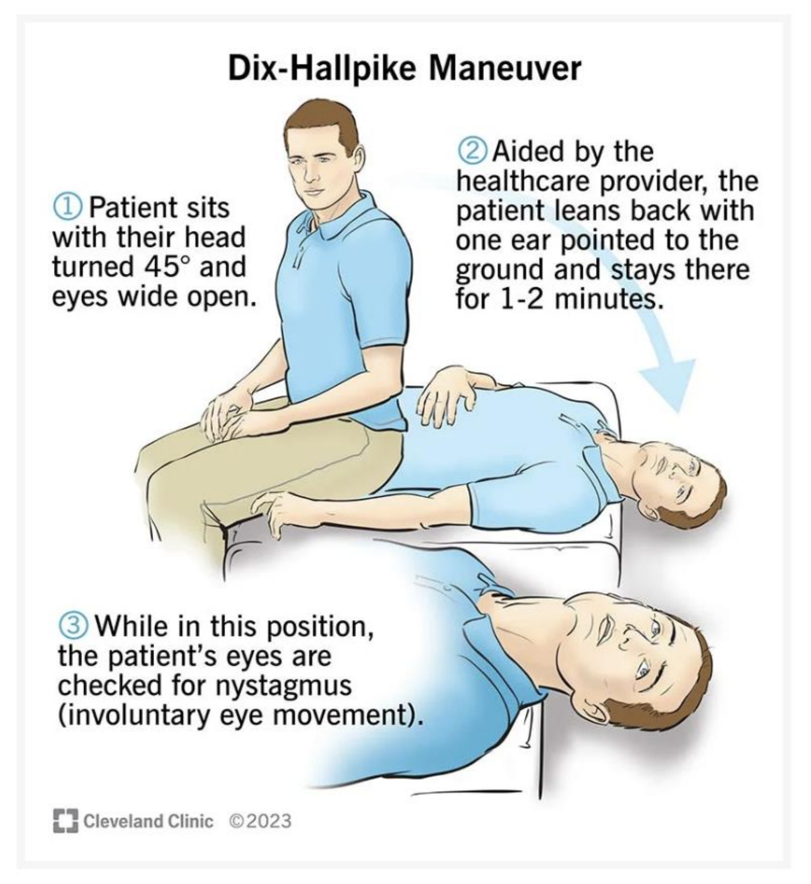
- Why was the VOR normal?
  - BPPV involves the semi-circular canals and the VOR tests the vestibular nerve
  - Canals really aren't involved in the VOR
- Dix Hallpike was positive because it does involve the semi-circular canals
- Keep in mind that Dix Hallpike will also be ***positive*** in vestibulitis, as the nerve cannot carry accurate information
- The patient will not appreciate a trial of Epley Maneuver with vestibulitis

Benign Paroxysmal Positional Vertigo



https://my.clevelandclinic.org/health/treatments/24859-dix-hallpike-maneuver

Clinical Case #5

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- Vitals: slightly elevated pulse rate, however remainder normal
- Nystagmus: yes
- Vestibular testing:
  - Fixations: extinguishes
  - VOR: abnormal
  - Postural stability: poor with eyes closed, okay with eyes open
  - Hearing: decreased on right
- Diagnosis?

Clinical Case #5

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- Vitals: slightly elevated pulse rate, however remainder normal

- Nystagmus: yes

- Vestibular testing:
  - Fixations: extinguishes
  - VOR: abnormal
  - Postural stability: poor with eyes closed, okay with eyes open
  - Hearing: decreased on right

- Diagnosis?
  - Meniere’s Disease

Meniere's Disease

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- Meniere's Disease
  - Classic triad of episodic vertigo, tinnitus, and hearing loss, likely caused by endolymphatic hydrops of the labyrinthine system of the inner ear

- Meniere's Syndrome
  - Symptomatic triad of episodic vertigo, tinnitus, and hearing loss occurring secondary to other ear disorders

- Onset typically between 20 to 40 years of age (however may occur at any age)

- Typically symptoms are present for about 3-5 years prior to diagnoses (unfortunately)

- Tests for Meniere's:
  - Audiometry, vestibular testing, MRI/CT of temporal bones

Meniere’s Disease

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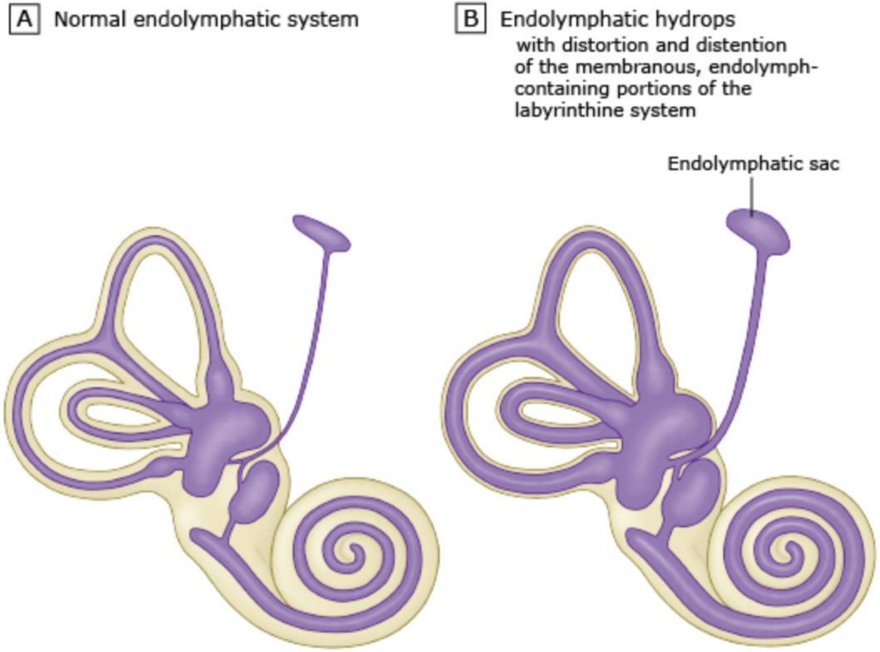

References:

1. *Meniere's disease: What is it? Harvard Health Publishing. Available at: https://www.health.harvard.edu/a_to_z/menieres-disease-a-to-z (Accessed on December 4,*

Meniere’s Disease

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Differential diagnosis

- Acoustic neuroma
- Multiple sclerosis
- Vascular loop compression of cranial nerve VIII
- Basilar/vertebral artery insufficiency
- Arnold-Chiari malformation
- Cerebellar tumors
- Transient ischemic attacks
- Benign paroxysmal positional vertigo
- Syphilitic endolymphatic hydrops
- Post-concussive hydrops
- Post-infectious hydrops (chronic otitis media, labyrinthitis)
- Autoimmune inner ear disease
- Perilymphatic Fistula
- Otosclerosis
- Migraine induced vertigo
- Diabetes
- Thyroid disease
- Cogan’s Syndrome
- Anemia

Important Points

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- Symptoms less than 1 minute –> BPPV
- Symptoms for hours –> Meniere’s or TIA/CVA
- Symptoms for days –> vestibulitis, mass, or CVA
- Inability to walk –> central (CVA)
- Does not extinguish –> central (CVA)
- VOR Abnormal – peripheral vertigo
- Dix Hallpike can be positive in more than BPPV
- History, history, history (important!)

Auditory Pathology (Basics)

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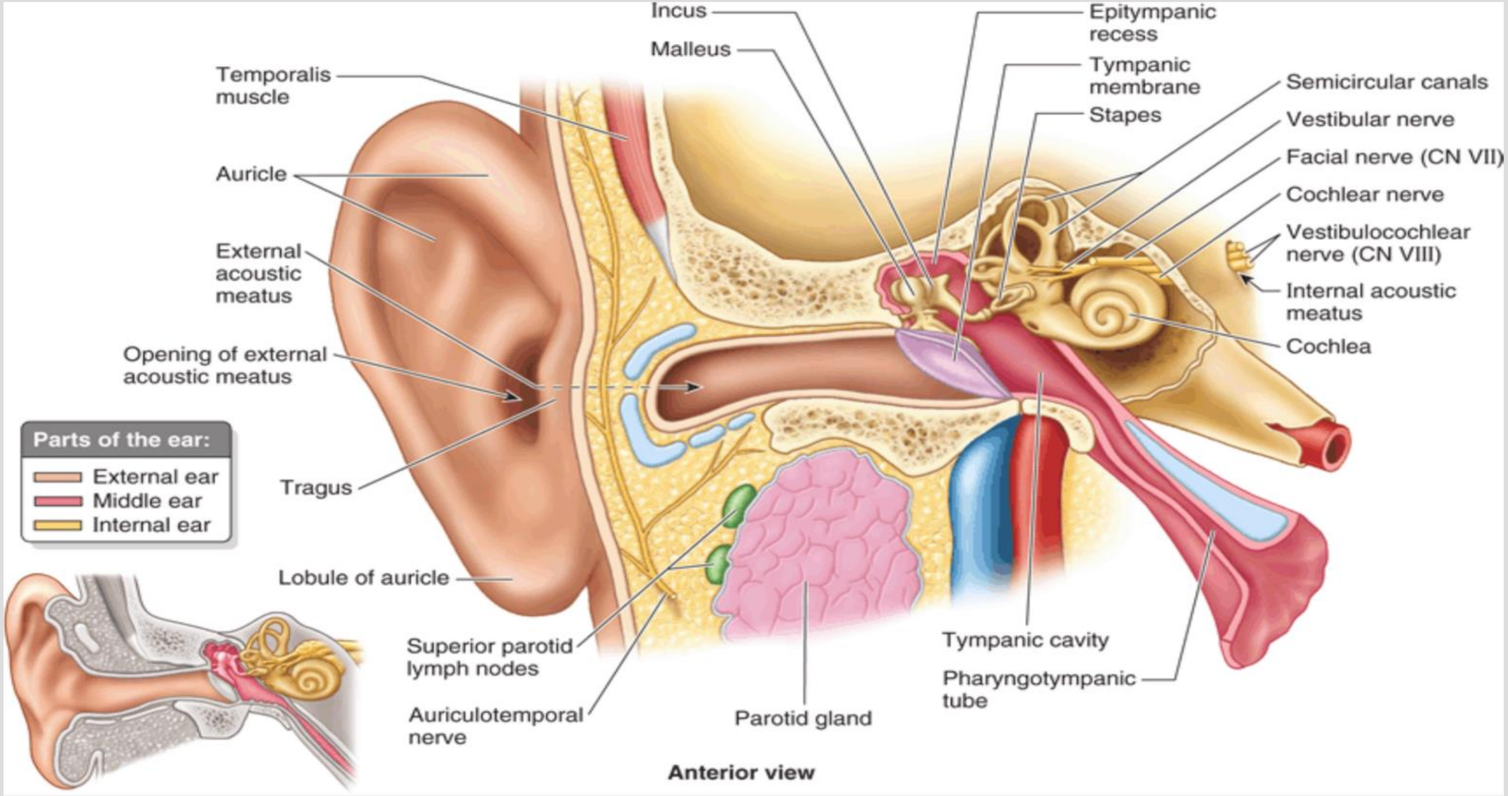
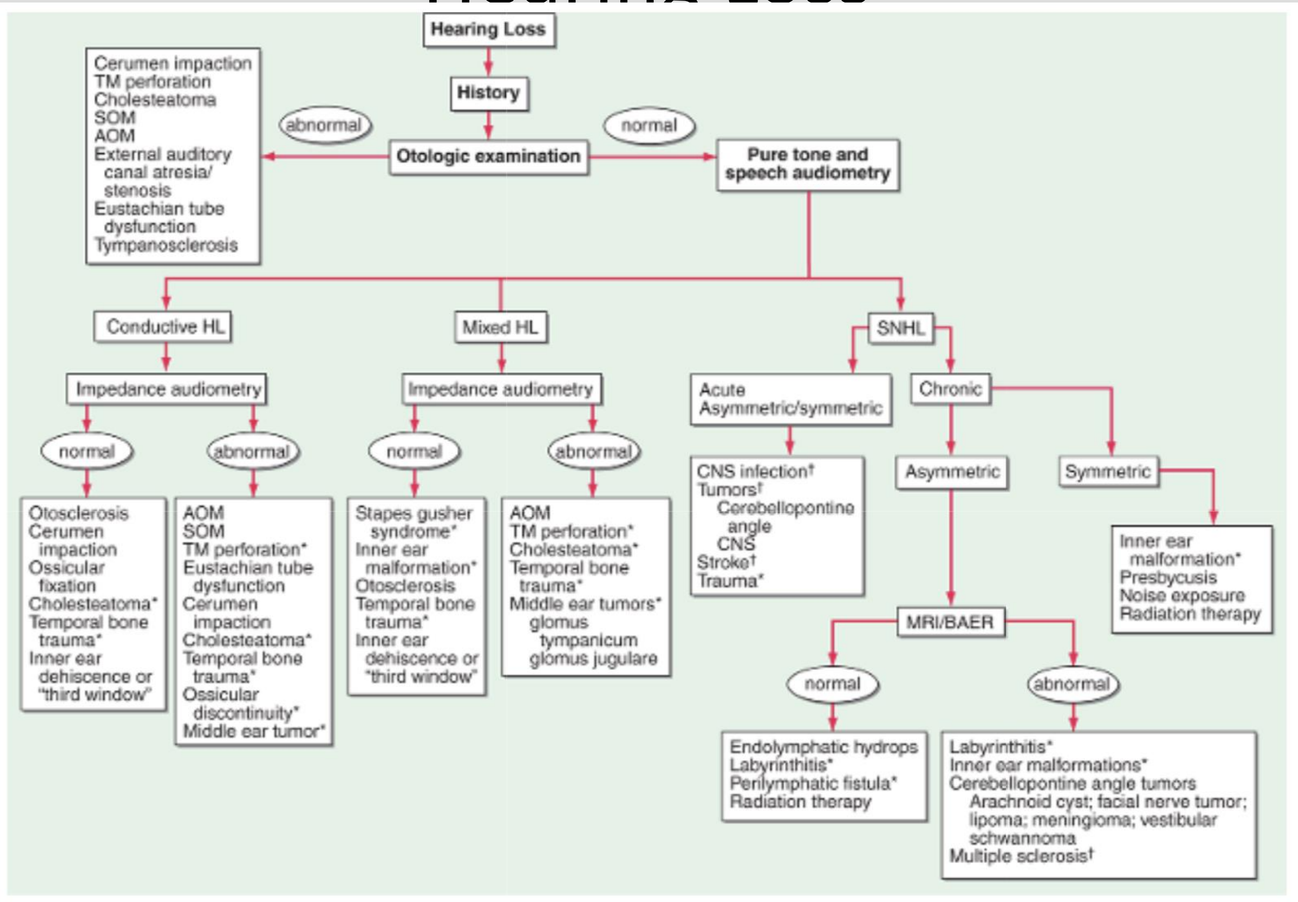


Moore's Clinically Oriented Anatomy, 7e, 2013

Hearing Loss



Harrison's/principles-of-internal-medicine-dan-longo-18th/chapter-30/figure-30-2

Hearing Loss

TABLE 30-1 Hereditary Hearing Impairment Genes

Designation	Gene	Function
Autosomal Dominant		
	CRYM	Thyroid hormone–binding protein
DFNA1	*DIAPH1*	Cytoskeletal protein
DFNA2A	*KCNQ4*	Potassium channel
DFNA2B	*GJB3 (Cx31)*	Gap junction
DFNA3A	*GJB2 (Cx26)*	Gap junction
DFNA3B	*GJB6 (Cx30)*	Gap junction
DFNA4	*MYH14*	Class II nonmuscle myosin
DFNA5	*DFNA5*	Unknown
DFNA6/14/38	*WFS1*	Transmembrane protein
DFNA8/12	*TECTA*	Tectorial membrane protein
DFNA9	*COCH*	Unknown
DFNA10	*EYA4*	Developmental gene
DFNA11	*MYO7A*	Cytoskeletal protein
DFNA13	*COL11A2*	Cytoskeletal protein
DFNA15	*POU4F3*	Transcription factor
DFNA17	*MYH9*	Cytoskeletal protein
DFNA20/26	*ACTG1*	Cytoskeletal protein
DFNA22	*MYO6*	Unconventional myosin
DFNA28	*TFCP2L3*	Transcription factor
DFNA36	*TMC1*	Transmembrane protein
DFNA44	*CCDC50*	Effector of EGF-mediated signaling
DFNA48	*MYO1A*	Unconventional myosin
DFNA50	*MIRN96*	MicroRNA
DFNA51	*TJP2*	Tight junction protein
Autosomal Recessive		
DFNB1A	*GJB2 (CX26)*	Gap junction
DFNB1B	*GJB6 (CX30)*	Gap junction
DFNB2	*MYO7A*	Cytoskeletal protein
DFNB3	*MYO15*	Cytoskeletal protein
DFNB4	*PDS(SLC26A4)*	Chloride/iodide transporter
DFNB6	*TMIE*	Transmembrane protein
DFNB7/B11	*TMC1*	Transmembrane protein
DFNB9	*OTOF*	Trafficking of membrane vesicles
DFNB8/10	*TMPRSS3*	Transmembrane serine protease
DFNB12	*CDH23*	Intercellular adherence protein
DFNB16	*STRC*	Stereocilia protein
DFNB18	*USH1C*	Unknown
DFNB21	*TECTA*	Tectorial membrane protein
DFNB22	*OTOA*	Gel attachment to nonsensory cell
DFNB23	*PCDH15*	Morphogenesis and cohesion
DFNB24	*RDX*	Cytoskeletal protein
DFNB25	*GRXCR1*	Reversible *S*-glutathionylation of proteins
DFNB28	*TRIOBP*	Cytoskeletal-organizing protein
DFNB29	*CLDN14*	Tight junctions
DFNB30	*MYO3A*	Hybrid motor-signaling myosin
DFNB31	*WHRN*	PDZ domain–containing protein
DFNB35	*ESRRB*	Estrogen-related receptor beta protein
DFNB36	*ESPN*	Ca-insensitive actin-bundling protein
DFNB37	*MYO6*	Unconventional myosin
DFNB39	*HFG*	Hepatocyte growth factor
DFNB49	*MARVELD2*	Tight junction protein
DFNB53	*COL11A2*	Collagen protein
DFNB59	*PJVK*	Zn-binding protein
DFNB61	*SLC26A5*	Motor protein
DFNB63	*LRTOMT/COMT2*	Putative methyltransferase
DFNB66/67	*LHFPL5*	Tetraspan protein
DFNB77	*LOXHD1*	Stereociliary protein
DFNB79	*TPRN*	Unknown
DFNB82	*GPSM2*	G protein signaling modulator
DFNB84	*PTPRQ*	Type III receptor-like protein-tyrosine phosphatase family

TABLE 30-2 Syndromic Hereditary Hearing Impairment Genes

Syndrome	Gene	Function
Alport syndrome	*COL4A3-5*	Cytoskeletal protein
BOR syndrome	*EYA1*	Developmental gene
	SIX5	Developmental gene
	SIX1	Developmental gene
Jervell and Lange-Nielsen syndrome	*KCNQ1*	Delayed rectifier K^+ channel
	KCNE1	Delayed rectifier K^+ channel
Norrie disease	*NDP*	Cell-cell interactions
Pendred syndrome	*SLC26A4*	Chloride/iodide transporter
	FOXI1	Transcriptional activator of *SLC26A4*
Treacher Collins	*TCOF1*	Nucleolar-cytoplasmic transport
Usher syndrome	*MYO7A*	Cytoskeletal protein
	USH1C	Unknown
	CDH23	Intercellular adherence protein
	PCDH15	Cell adhesion molecule
	SANS	Harmonin-associated protein
	USH2A	Cell adhesion molecule
	VLGR1	G protein–coupled receptor
	USH3	Unknown
	WHRN	PDZ domain–containing protein
WS type I, III	*PAX3*	Transcription factor
WS type II	*MITF*	Transcription factor
	SNAI2	Transcription factor
WS type IV	*EDNRB*	Endothelin B receptor
	EDN3	Endothelin B receptor ligand
	SOX10	Transcription factor

Abbreviations: BOR, branchio-oto-renal syndrome; WS, Waardenburg syndrome.

Harrison's/principles-of-internal-medicine-dan-longo-18th/chapter-30/tables-30-1&2

Evaluation of Hearing Loss

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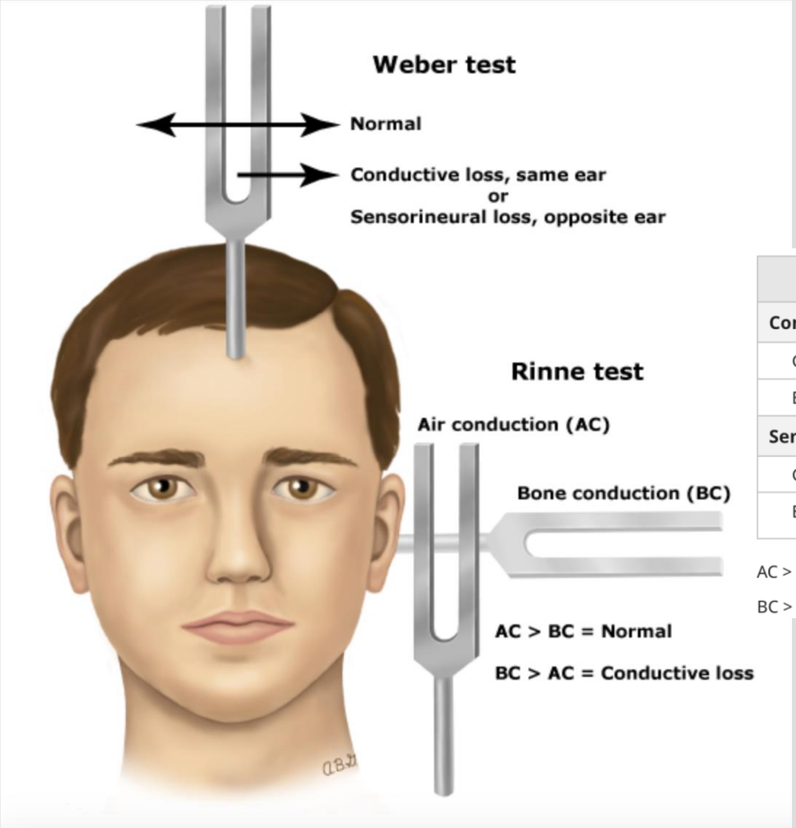
- Goal of evaluation:
  - Determine nature of the hearing impairment (sensorineural or conductive)
  - Severity of the impairment
  - Anatomy (location) of the impairment
  - Etiology
- History!
- Physical Exam
  - External to internal (try to avoid irrigation of ears if possible)
- Special testing
  - Rinne and Weber tuning fork tests (512-Hz tuning fork)
- Audiological assessment
- Pneumoscopy
  - Nonmobile TM -> fluid or mass in middle ear cavity, or a stiff/sclerotic TM
  - Hypermobile TM -> ossicular chain disruption
  - Only moves with negative pressure -> blocked eustachian tube
- Temporal bone CT scan, MRI with gadolinium (if needed)
- Laboratory analysis
  - Blood glucose, CBC with differential, thyroid studies, RPR, ANA profile

Weber and Rinne



	Weber lateralizes	Rinne test
Conductive loss		
Good ear		AC > BC
Bad ear	To bad ear	BC > AC
Sensorineural loss		
Good ear	To good ear	AC > BC
Bad ear		AC > BC*

AC > BC: Air conduction better than bone conduction (normal Rinne).

BC > AC: Bone conduction better than air conduction (abnormal Rinne).

https://www.uptodate.com/contents/evaluation-of-hearing-loss-in-adults?search=disorders%20of%20hearing&source=search_result&selectedTitle=7~150&usage_type=default&display_rank=7#H6

Weber and Rinne

- Must use results from **BOTH** Weber and Rinne testing!

- Example #1:
  - Weber lateralized to RIGHT; Rinne positive BILATERALLY
  - Weber (may mean conductive hearing loss on right OR sensorineural loss on the left)
  - Rinne (AC > BC both sides, so no conductive hearing loss)
  - This suggests Left sided Sensorineural loss

Tinnitus

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- Estimated 50 million people in the US alone have chronic tinnitus
- Subjective vs objective
- Theories
  - Loss of cochlear input to the central auditory system
  - Loss of suppressive neural connections (as seen in phantom pain)
  - Abnormalities in serotonin levels in the auditory cortex causing aberrant neural firing
- Multiple etiologies
  - Ototoxicity, hypertension, diabetes, bone disease, stroke, tumors, MS, medications, etc.
- Patients to be referred to ENT when tinnitus is pulsatile, unilateral, sudden hearing loss, dizziness, or ear/facial pain; otherwise, further testing not needed in most patients
- Treatment
  - Remove/treat offending cause, CBT, masking, electrical stimulation of cochlea

Important Points

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- History is vastly important
- Conductive vs Sensorineural
- Weber and Rinne testing must be interpreted together
- Further testing or imaging may be required